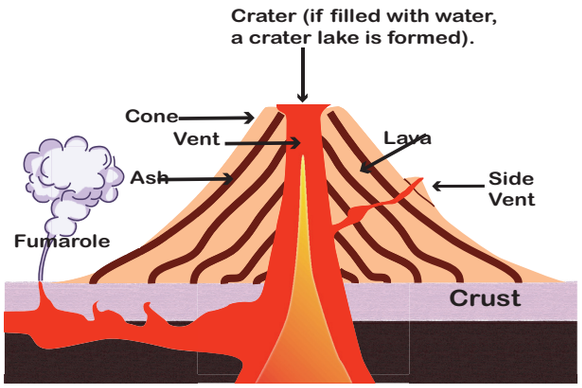


Figure 3.25 Volcano

Caldera. Volcanic ash consists of fragments of pulverized rock, minerals and volcanic glass, created during volcanic eruptions. Volcano generally erupts either through the vent (E.g. Mt. Fujiyama, Japan) or fissure (The Deccan Plateau, India).

Pumice is a volcanic rock produced when lava with a very high content of water and gases is discharged from a volcano.

3.10.1 Causes of Volcanic Eruptions

The following are the causes of volcanic eruptions:

Weak Zones in the Earth Crust: The parts of the earth where two tectonic plates collide against or drift apart from each other are considered very weak. Volcanoes may erupt in such zones, for example, African and Eurasian plates.

Magma Saturated with Gases: The magma, in the interior of the earth, is often found saturated with gases like carbon dioxide, and hydrogen sulphide. These gases together with water vapour make the magma highly explosive. Magma is forced out as lava on the surface of the earth due to the pressure exerted by these gases.

3.10.2 Types of Volcanoes

Based on the **frequency of eruption**, there are three types of volcanoes:

Fact File

Mt. Krakatau

The greatest volcanic explosion known to humans is perhaps Mt. Krakatau in August 1883.

Krakatau is a small volcanic island in the Sunda Straits, between Java and Sumatra.

The explosion could be heard in Australia, almost 4,000 km away.

The vibration set up enormous waves over 30 m high which drowned 36,000 people in the coastal districts of Indonesia.



Cotopaxi in Ecuador is the world's highest active volcano.

1. **Active Volcanoes:** Volcanoes which erupt frequently are called active volcanoes. Generally, their vent remains open. Mount Etna of Italy, Cotopaxi in Ecuador are some examples.
2. **Dormant Volcanoes:** These volcanoes may not have erupted in the recent past but there is a possibility of eruption at any time. In other words, they may lie dormant awaiting active eruption anytime. Sometimes gases and steam come out of them. They cause great destruction to life and property once they become active again. Mt. Vesuvius of Italy and Mt. Fujiyama of Japan are examples.
3. **Extinct Volcanoes:** These volcanoes have exhausted their energy and have

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